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Predicting the unpredictable? Lessons learned from FT-NIRS age prediction using diverse ageing structures

Michelle S. Passerotti
NOAA Fisheries Northeast Fisheries Science Center
Apex Predators Program



Michelle Passerotti
Ph.D. 2021, University of South Carolina

“Applications and mechanisms of near infrared spectroscopy on age estimation in otoliths of red snapper *Lutjanus campechanus*”

Research fish biologist specializing in life history studies of sharks and bony fishes

michelle.passerotti@noaa.gov

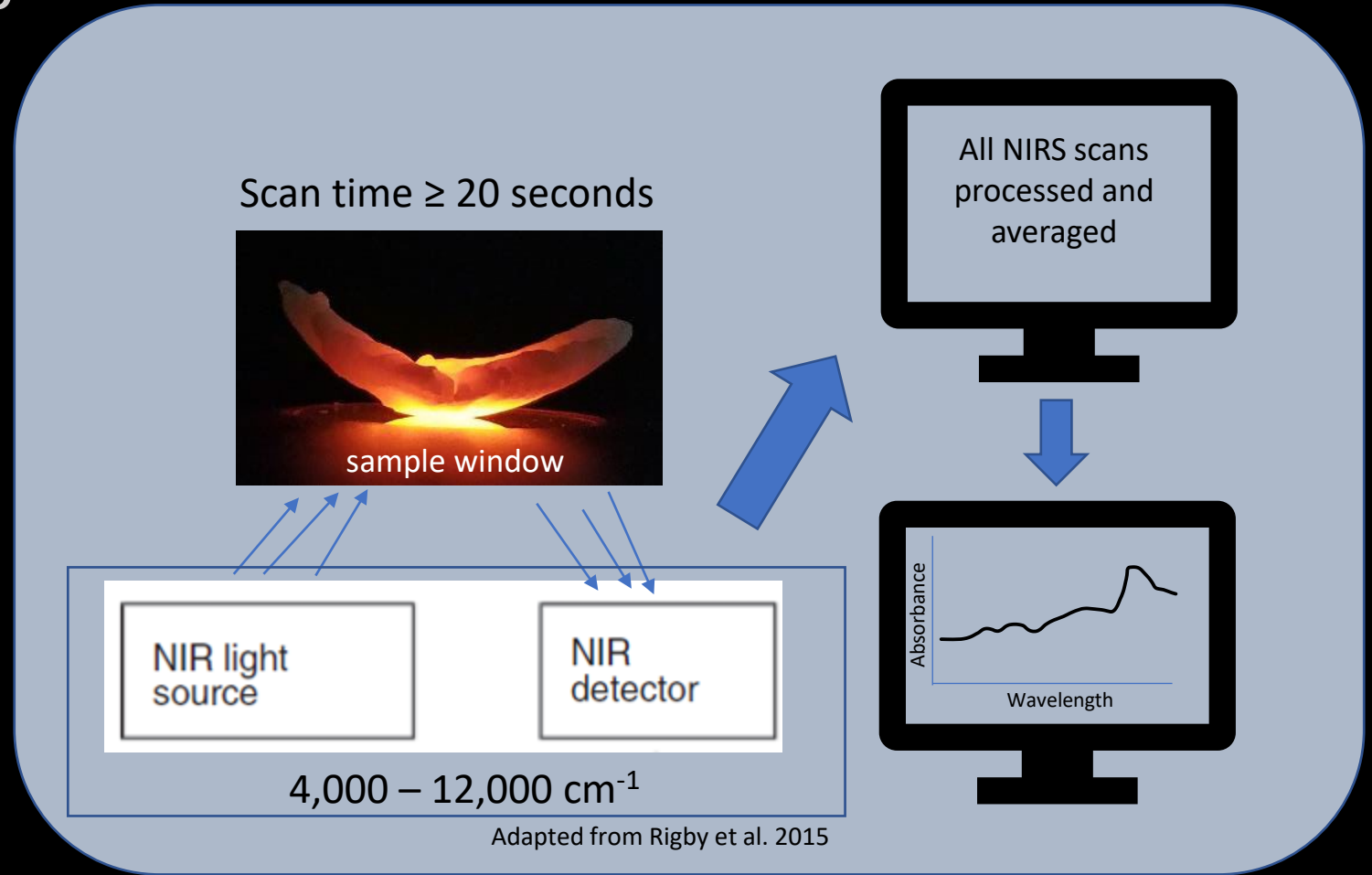
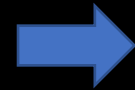


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FT-NIRS: What's in the box?

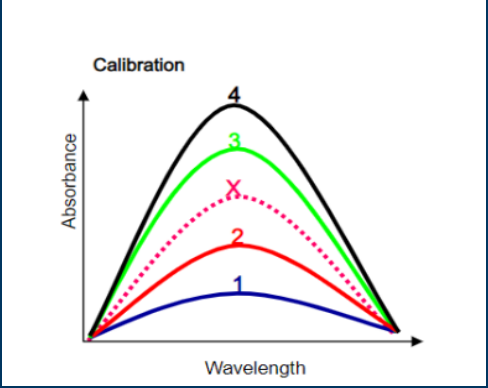
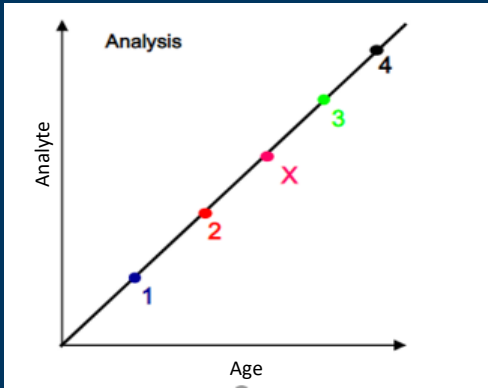


Bruker Matrix FT-NIR spectrometer



- Comparison of multiple samples across varying age classes to identify differences in spectral signatures
- Multivariate statistical analyses / “chemometrics” to form predictive models based upon wavelength-specific differences as related to age

Predicting the unpredictable? Mechanisms of NIRS age prediction



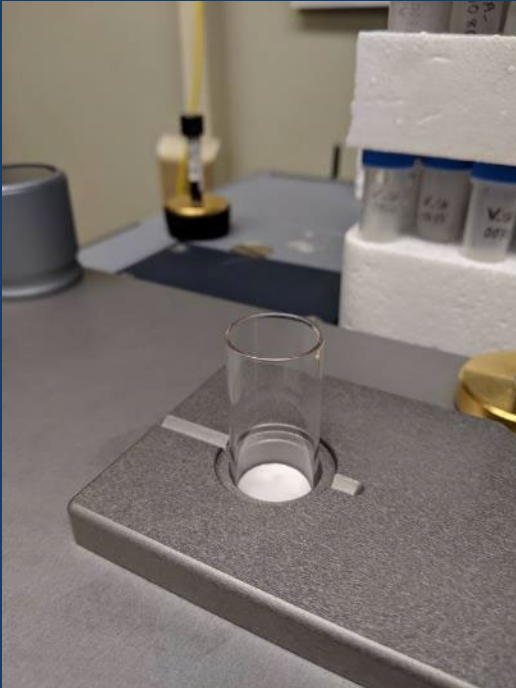
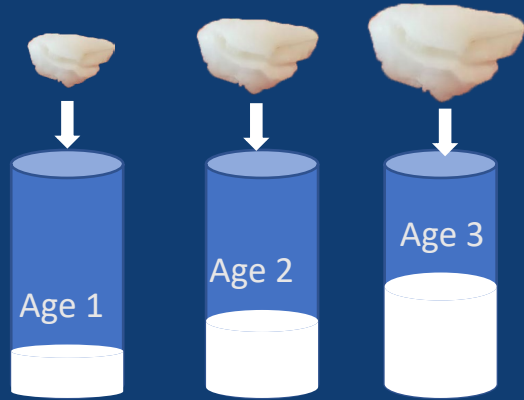
Physicochemical mechanisms of FT-NIRS age prediction in fish otoliths

Michelle S. Passerotti^{A,B,*} , Marcel J. M. Reichert^C, Bailey A. Robertory^{A,D}, Zachary Marsh^E, Morgan Stefik^E and Joseph M. Quattro^F

Age prediction: chemical or physical?

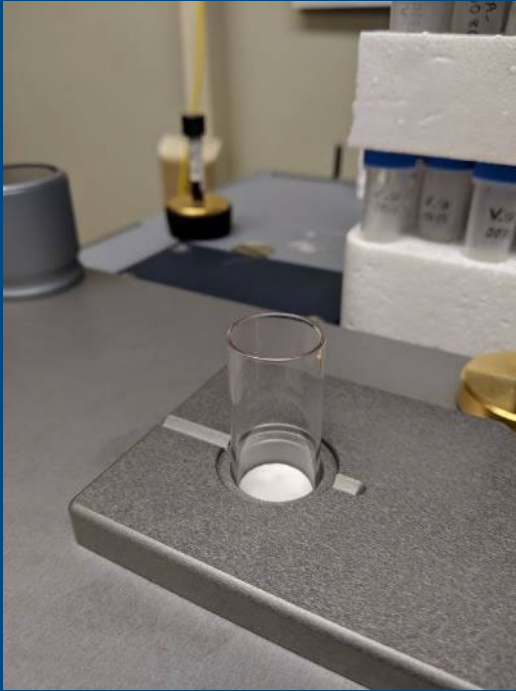
1. Do the inherent age-related structural changes in whole otoliths underlie age related spectral differences?
2. Do age-related changes in otolith mass underlie age-related spectral differences? What about changing mass without changing concentration?
3. Does otolith protein content change with age, and is it predictable using FT-NIRS?

Methods: Otolith structure



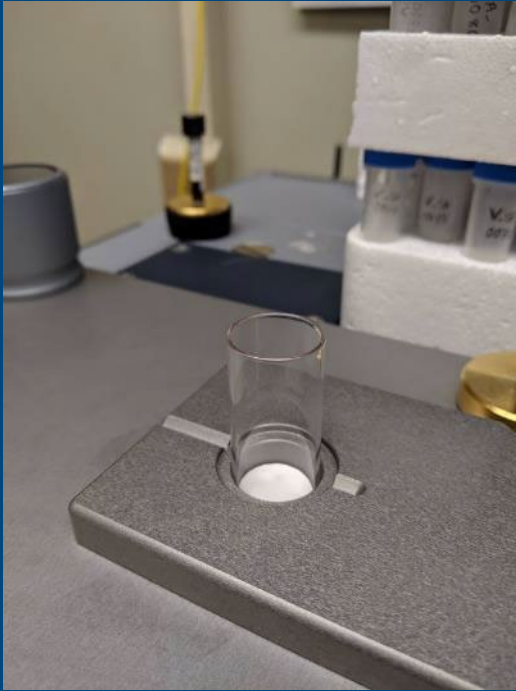
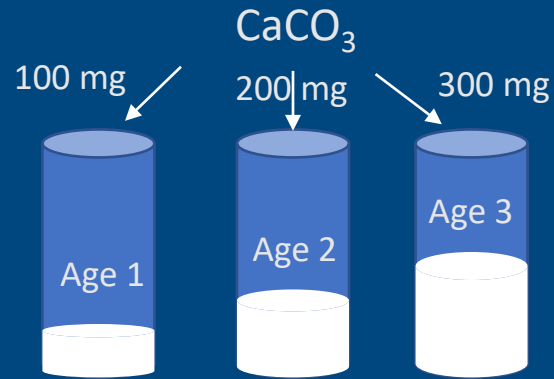
- Sample set of n=84 Atlantic otoliths ages 0 – 25 (stored dry)
 - Scanned whole
 - Ground otoliths to powder (~250 microns)
 - Poured full mass of otolith powder into glass cuvettes and scanned individually following whole otolith protocol
 - PLS age regression formulated as for whole otoliths:
 - Standardized preprocessing (1st derivative Savitsky Golay) and wavenumber regions (6104-4200 cm⁻¹) to facilitate comparison
 - Optimized wavenumber model

Methods: Otolith mass



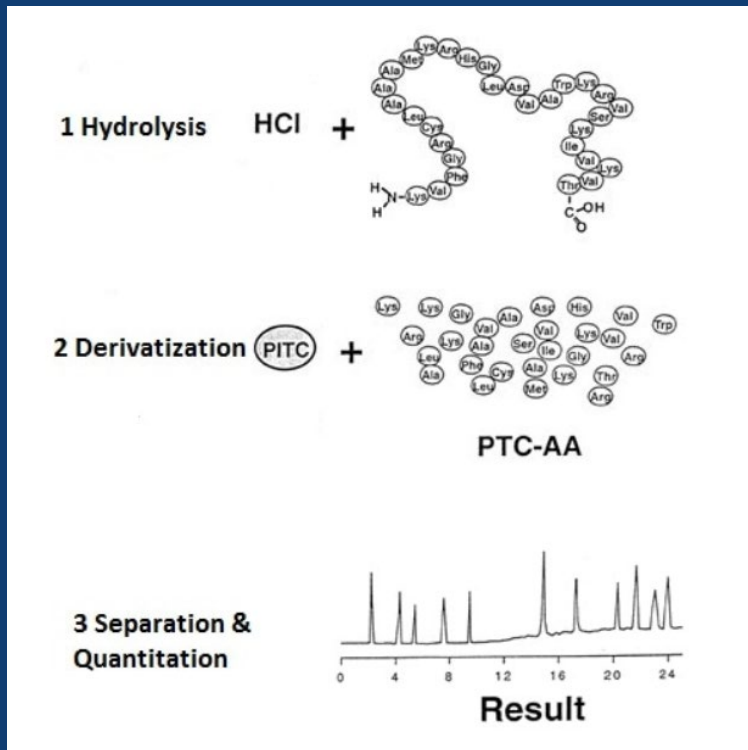
- $n=71$ ground otoliths (ages 3+), 300 mg of otolith powder was subsampled from each into clean cuvettes and scanned individually
- PLS age regression formulated as for whole otoliths
 - Standardized preprocessing and wavenumber regions to facilitate comparison
 - Optimized wavenumber model

Methods: Calcium carbonate mass



- Pure precipitated calcium carbonate (calcite; Amresco)
- 100 - 1000 mg quantities of CaCO_3 powder were measured in 100 mg intervals into clean cuvettes and scanned in triplicate
- Each mass was assigned a dummy variable “layer age” based on mass... 100 mg = 1 yr, 200 mg = 2 yr, etc.
- PLS age regression formulated as for whole otoliths
 - Standardized preprocessing and wavenumber regions to facilitate comparison
 - Optimized wavenumber model

Methods: Otolith protein content

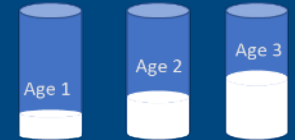
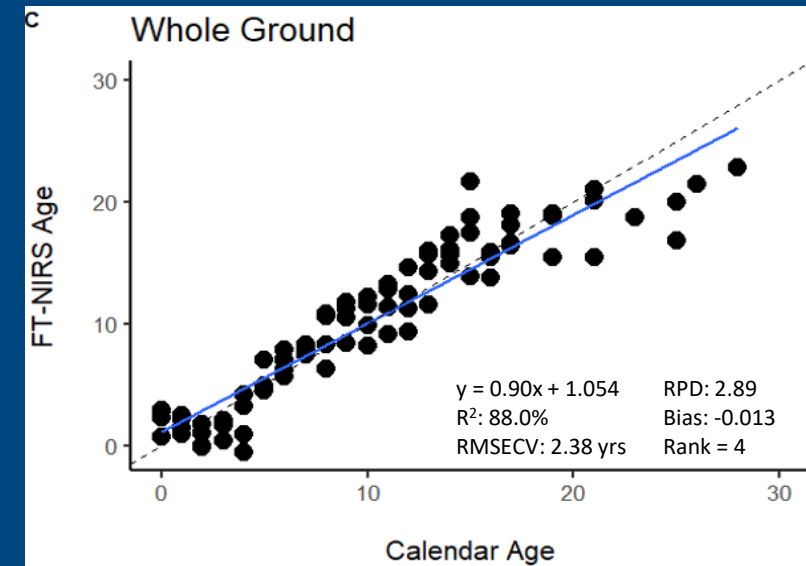
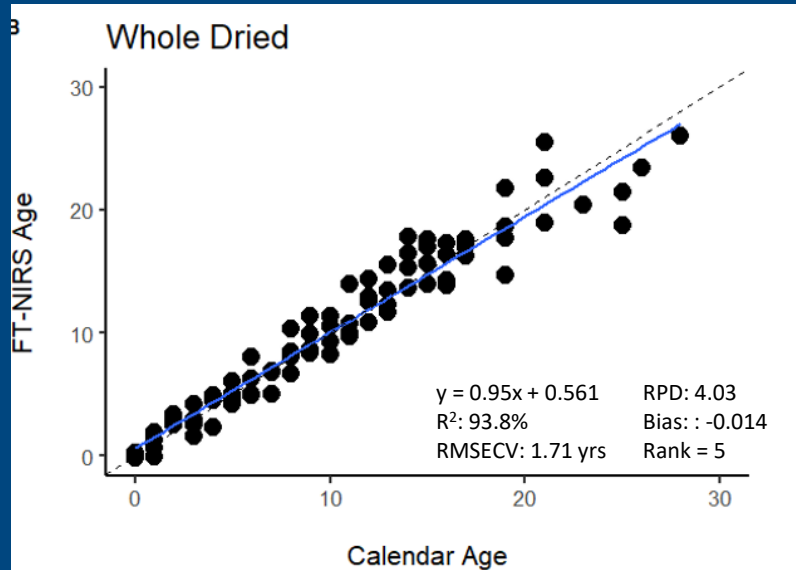


- N = 26 ground otoliths ranging 0-25 years old were subsampled for protein quantification
- UC Davis Genome Center – Molecular Structure Facility
 - Acid hydrolysis
 - 6N HCL, 1% Phenol, 110°C, 24hr, in vacuo
 - Ninhydrin detection, Hitachi 8800 analyzer
- Individual amino acid residues summed to calculate total protein concentration (% otolith weight)
- PLS protein prediction models formulated using spectra from both whole and ground otoliths
 - Standardized preprocessing and wavenumber regions to facilitate comparison
 - Optimized wavenumber model

Results: Age model comparison, standardized region (6104-4200 cm⁻¹)



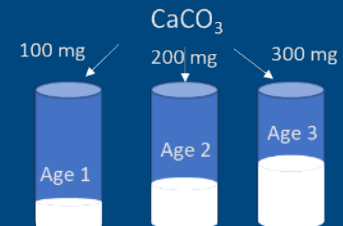
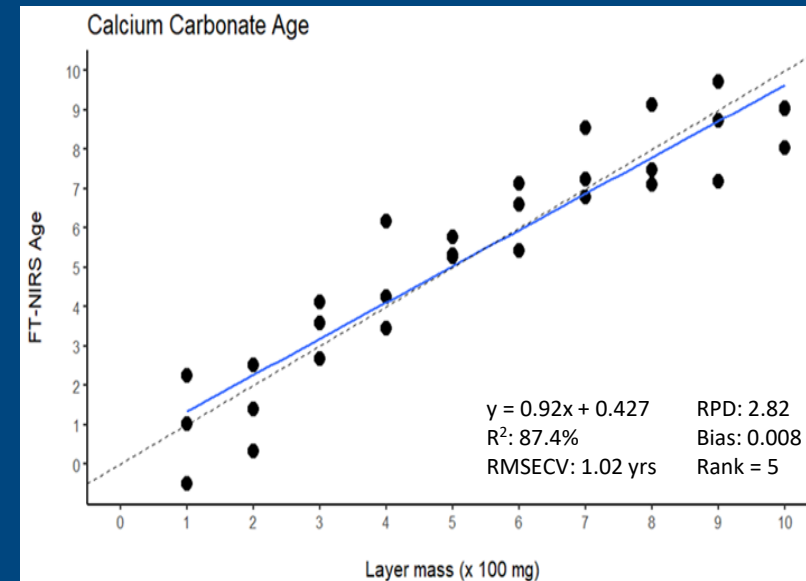
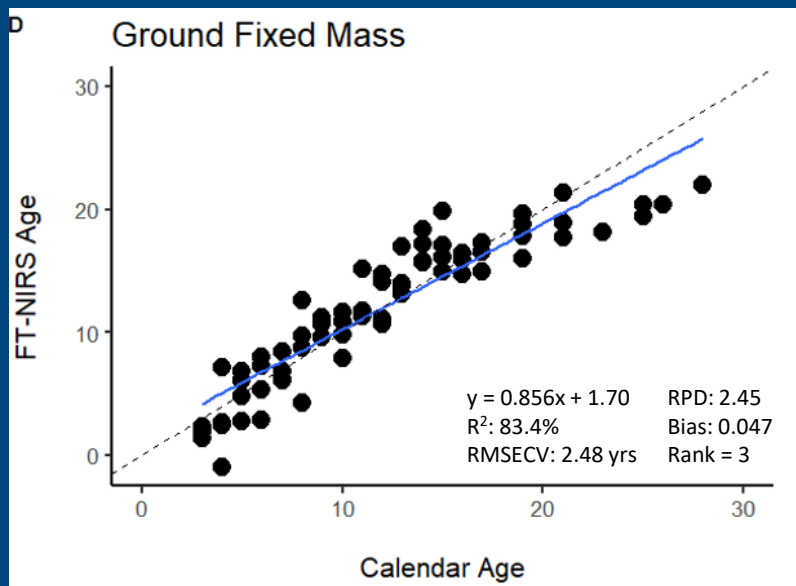
Optimized
R²: 94.4%
RMSECV: 1.62
RPD: 4.23



Optimized
R²: 89.9
RMSECV: 2.19
RPD: 3.15

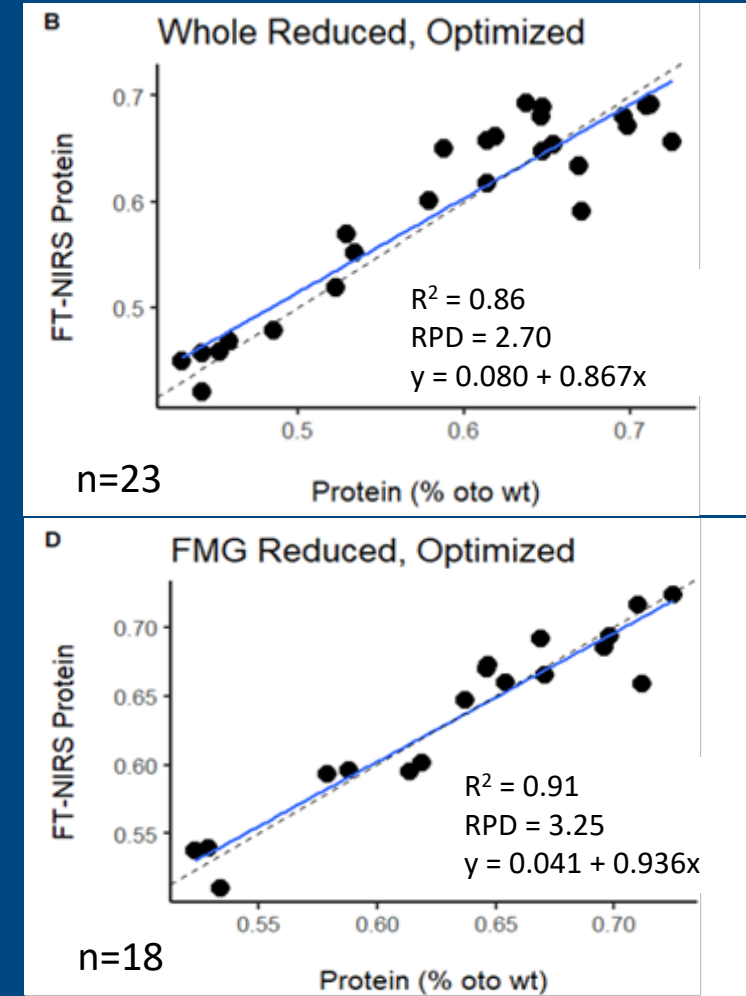
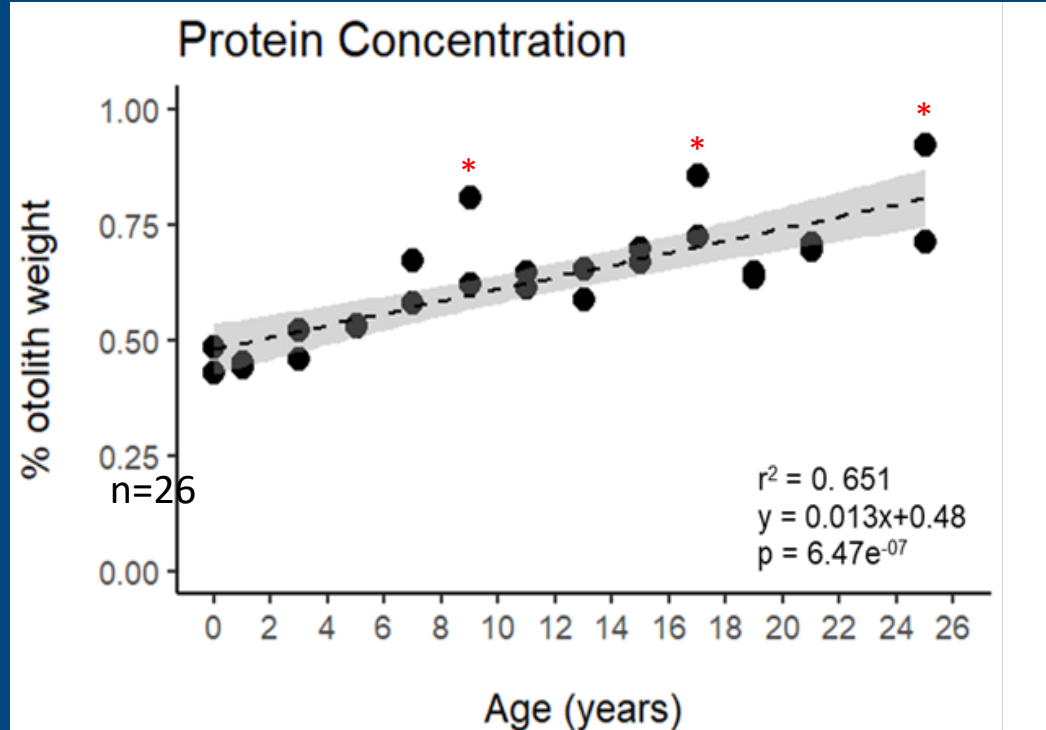


Optimized
R²: 89.0
RMSECV: 2.02
RPD: 3.02



Optimized
R²: 90.7
RMSECV: 0.878
RPD: 3.27

Results: Protein content and NIRS protein prediction



Conclusions

- 1. Structural changes in whole otoliths are a factor in age related spectral differences.
 - Structure (whole otolith) explained up to 11% of age model variance and reduced model error substantially
- 2. Age-related changes in otolith mass do not underlie spectral differences, but do contribute.
 - Mass alone explained about 5% of the age model variance in whole ground vs fixed-mass otoliths, age explained ~85%
 - Mass increase without concentration increase can be used to create false age correlation
- 3. Protein concentration increases with age, is detectable and predictable by NIRS at concentrations found in otoliths, and protein regions are included in age prediction models.

NIRS spectral signatures represent the sum total of physical and chemical properties of otoliths

Case study: Sand tiger shark vertebrae

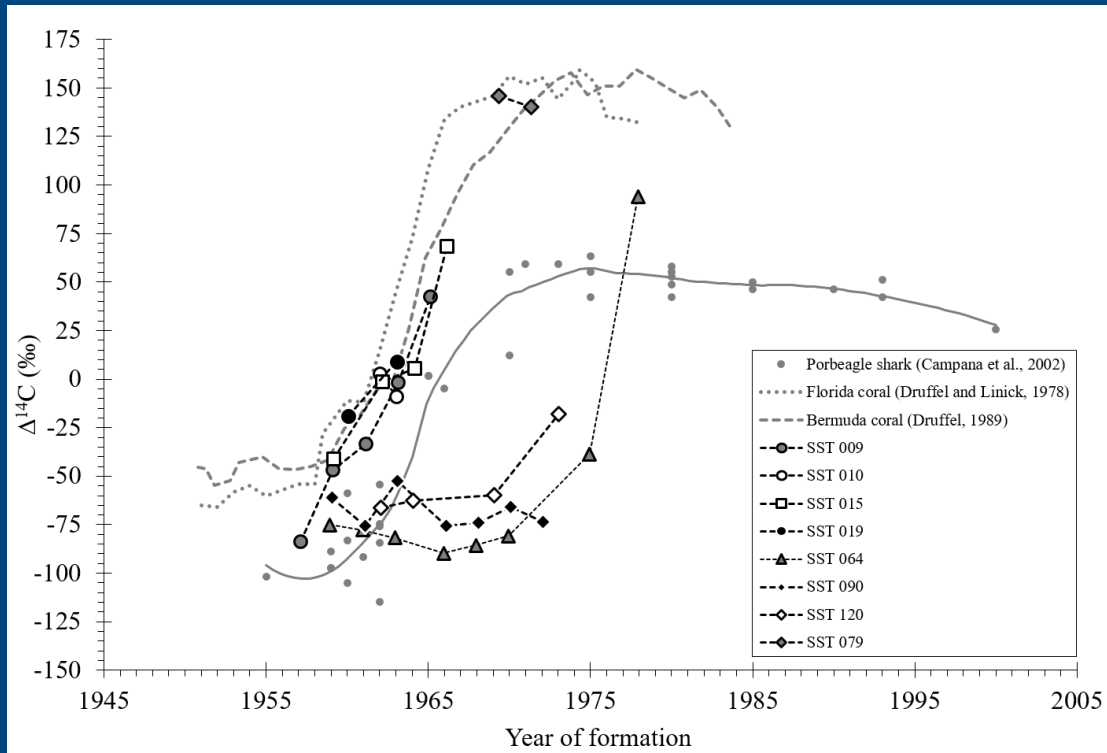
Missing time in annuli counts

Sand tiger shark vertebrae stop recording age accurately after ~17 years

Validated using bomb-radiocarbon analysis, no other ageing technique available



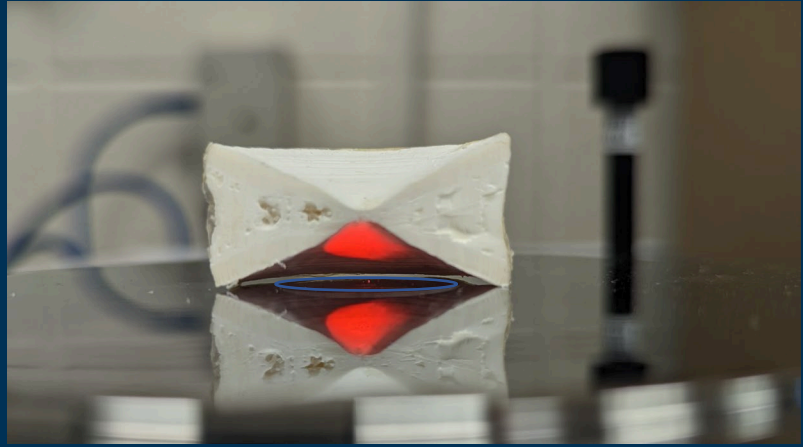
Sand tiger shark, *Carcharias taurus*



Passerotti et al. 2014

Does NIRS measure some age-related compound in vertebrae that is independent of growth band deposition?

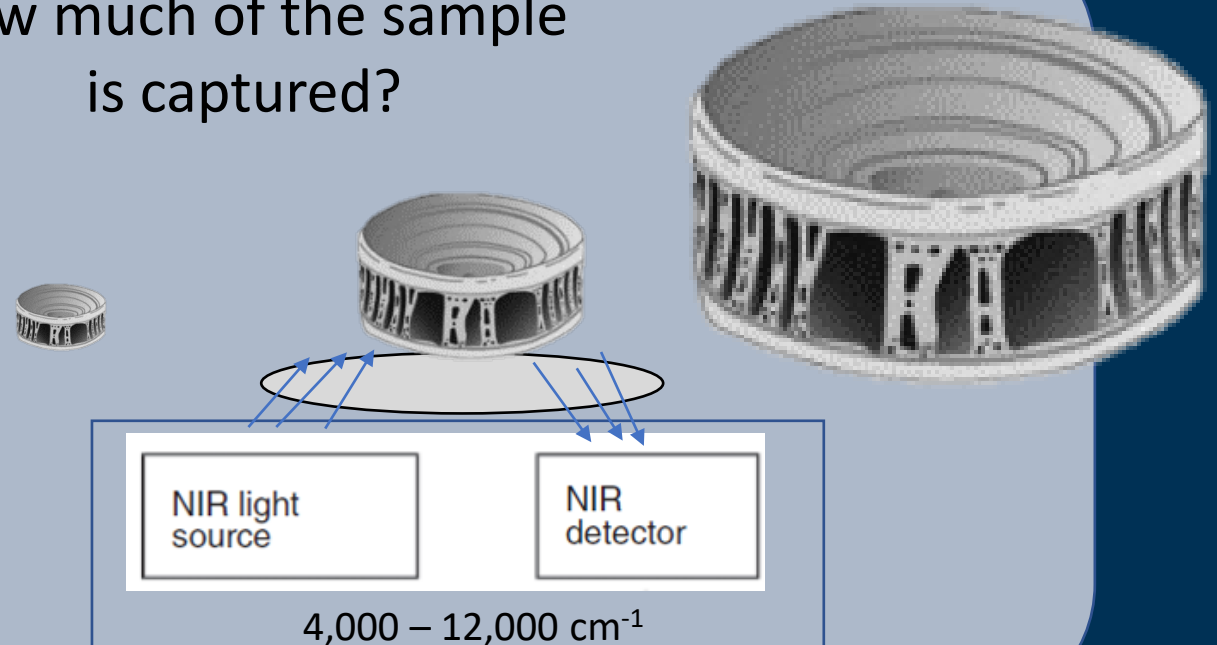
Sand tiger shark vertebrae: physical vs chemical changes?



Limiting factors?

- Size
- Missing growth bands

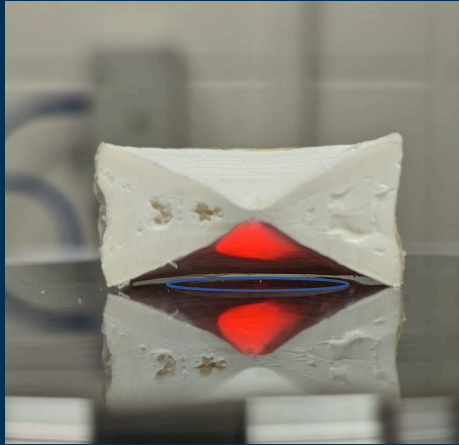
How much of the sample is captured?



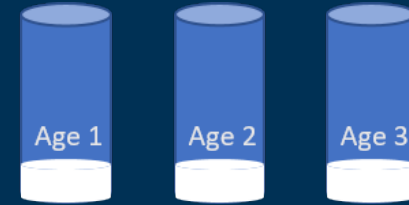
Adapted from Rigby et al. 2015

Vertebral analysis: how does morphology affect age prediction?

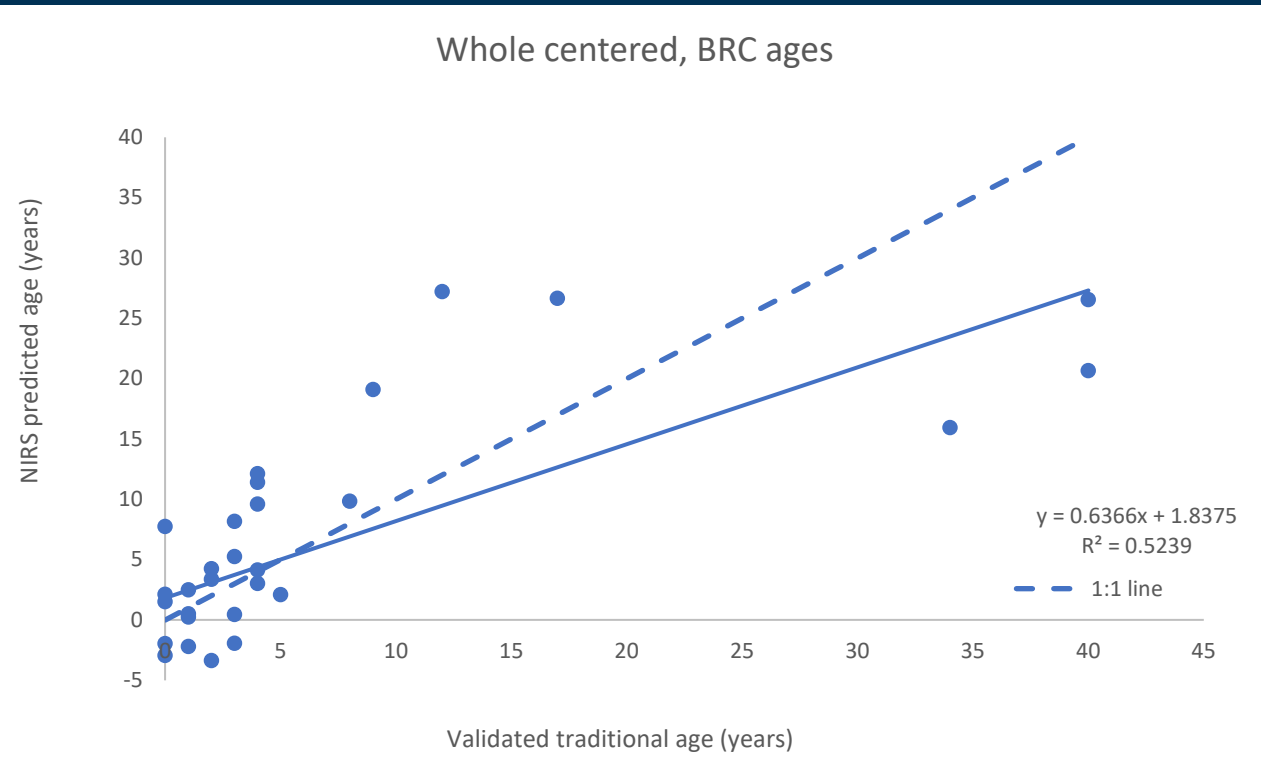
1. Whole vertebrae centered on window vs halved vertebrae scanned along growth axis



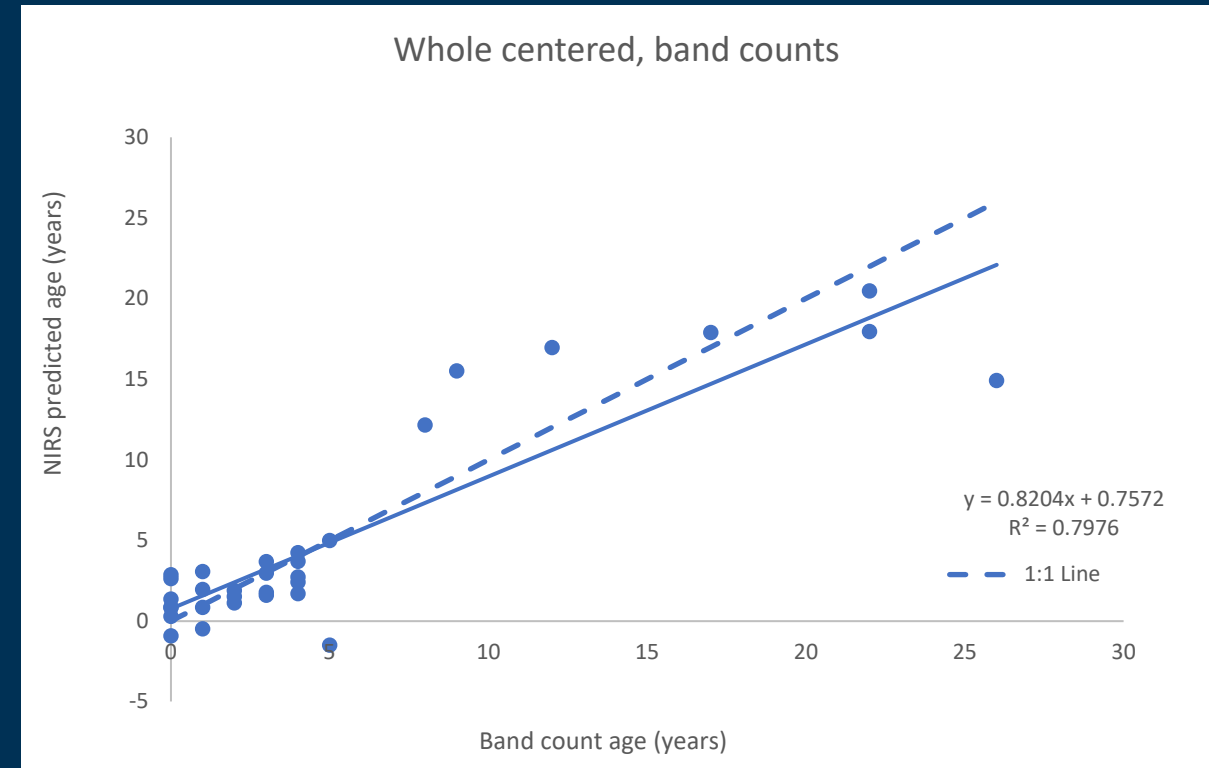
2. 300mg subsamples of powdered vertebrae



Preliminary Results: NIRS prediction from whole vertebrae

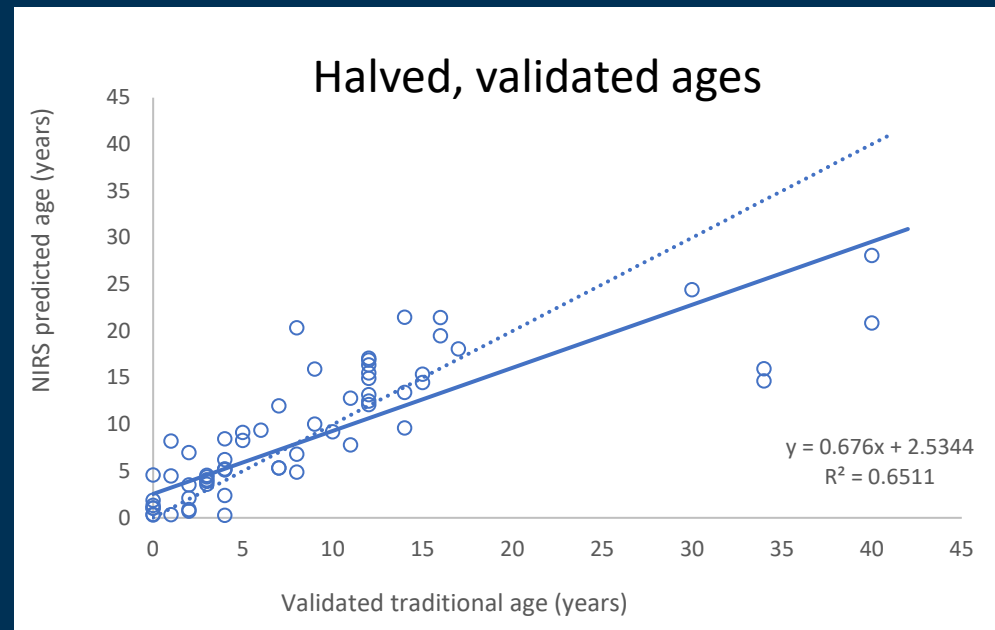
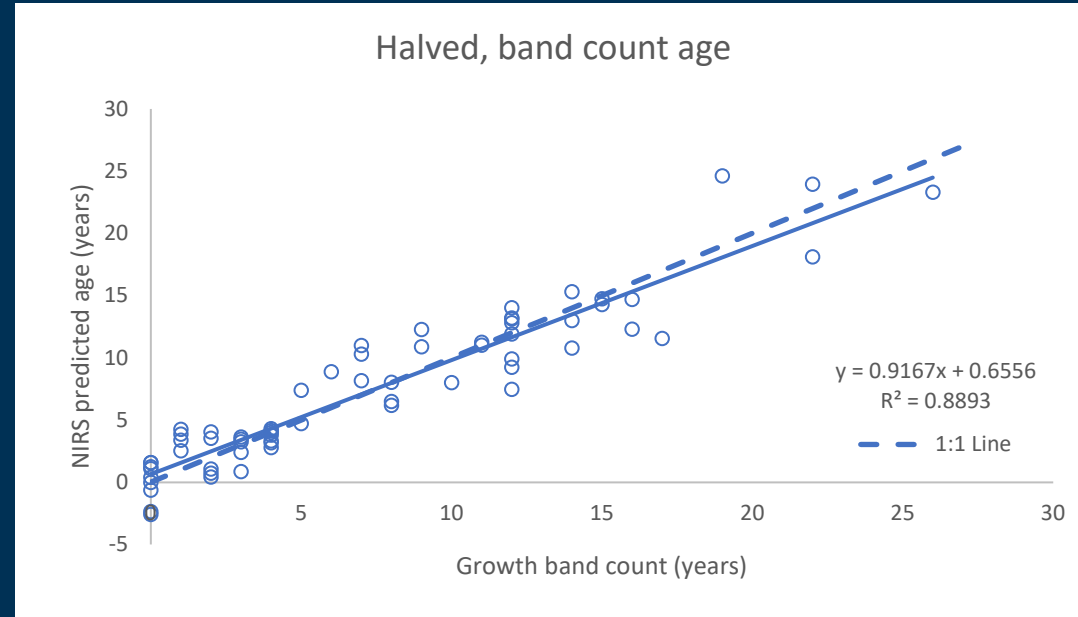
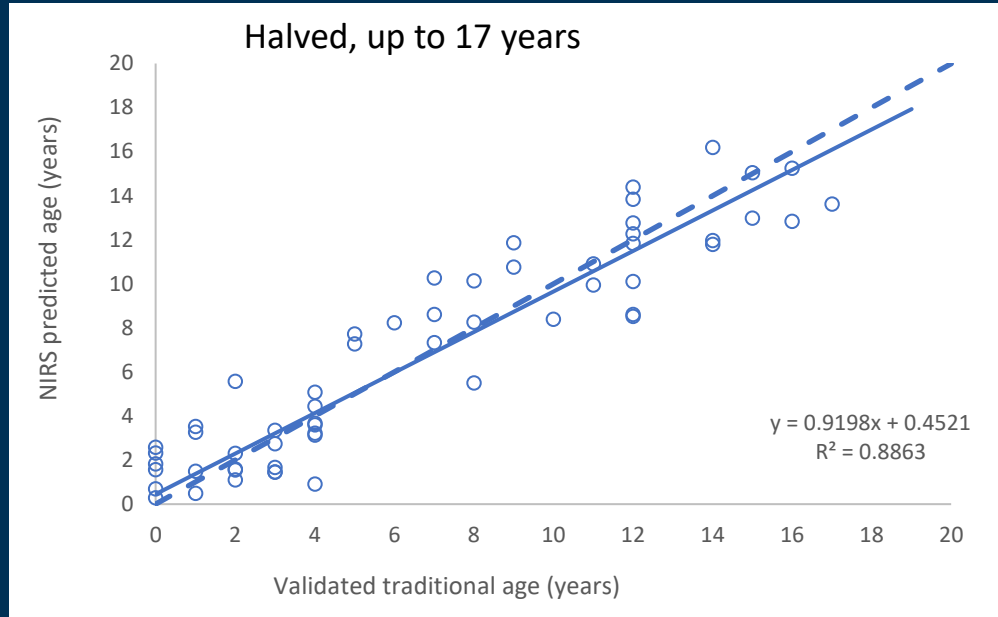


NIRS age error range: -15 to 19 years

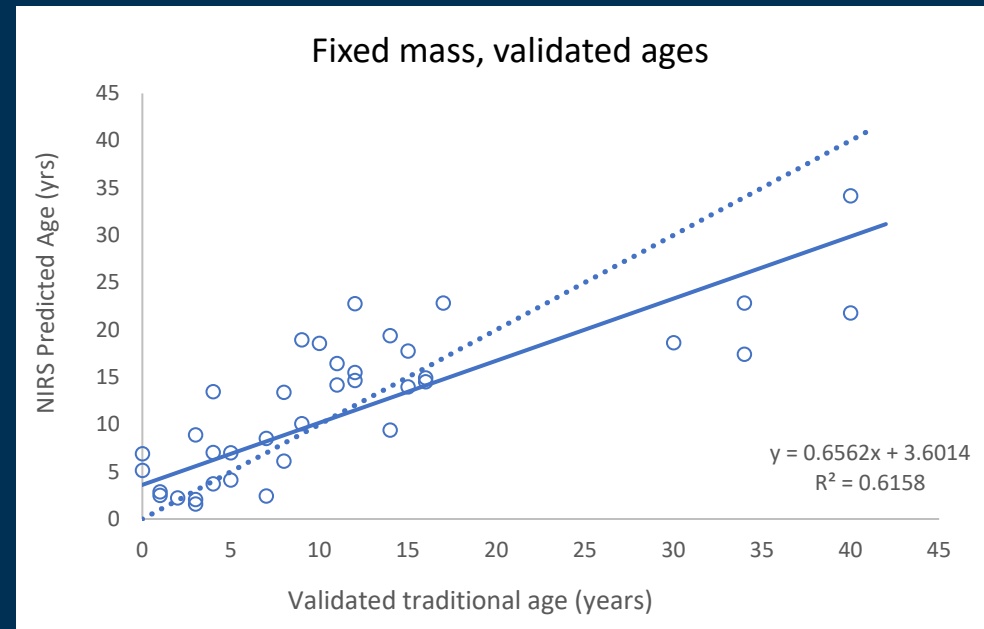
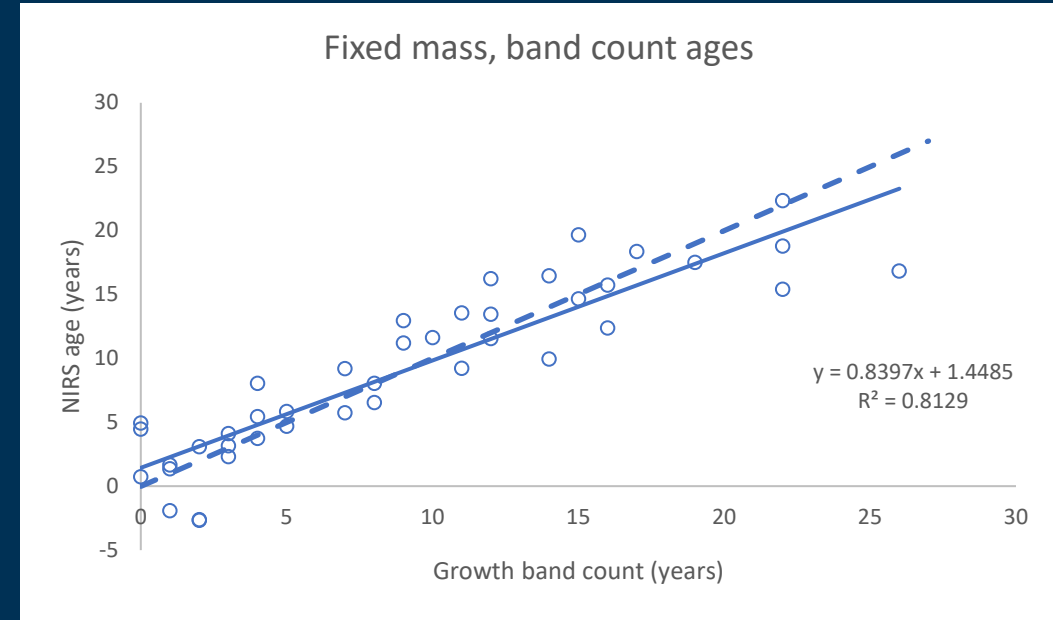
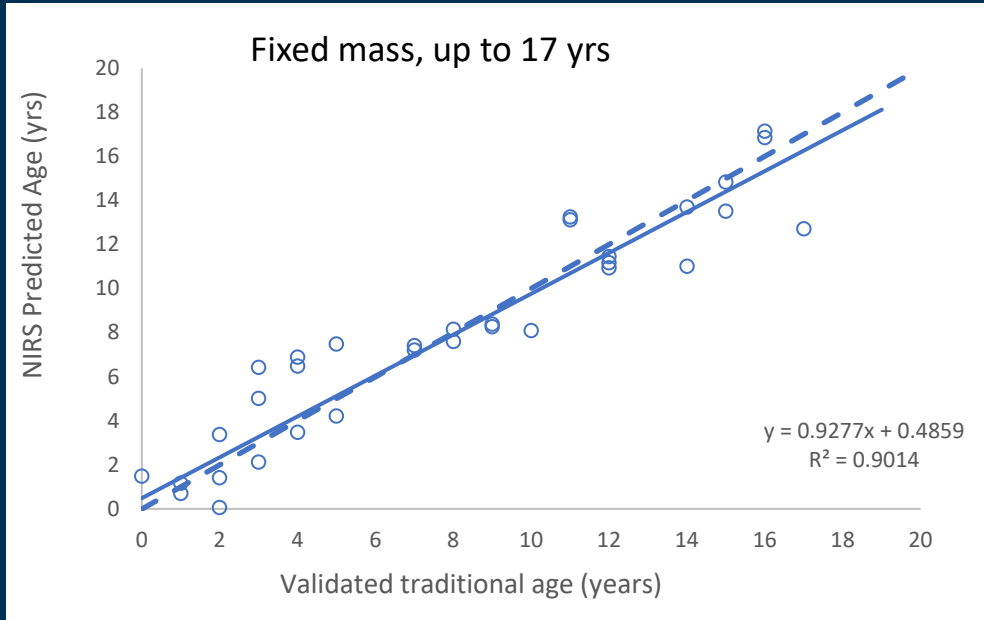


NIRS age error range: -6.5 to 11 years

Preliminary Results: NIRS prediction from halved vertebrae



Preliminary Results: NIRS prediction from powdered vertebrae

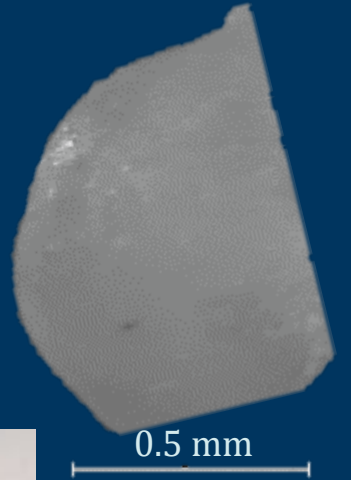


Conclusions: Case study #1

- 1. Preliminary evidence from shark vertebrae that NIRS relies on accreted material over time, not diagenetic/racemic change to existing material;
in sand tigers, reflects vertebral growth more than age
- 2. True chemical differences in this accreted material underlie NIRS age prediction capability
- 3. Differences in age structure morphology across species will challenge the use of NIRS for age prediction at a management scale

Case study: Northern shortfin squid *Illex illecebrosus*

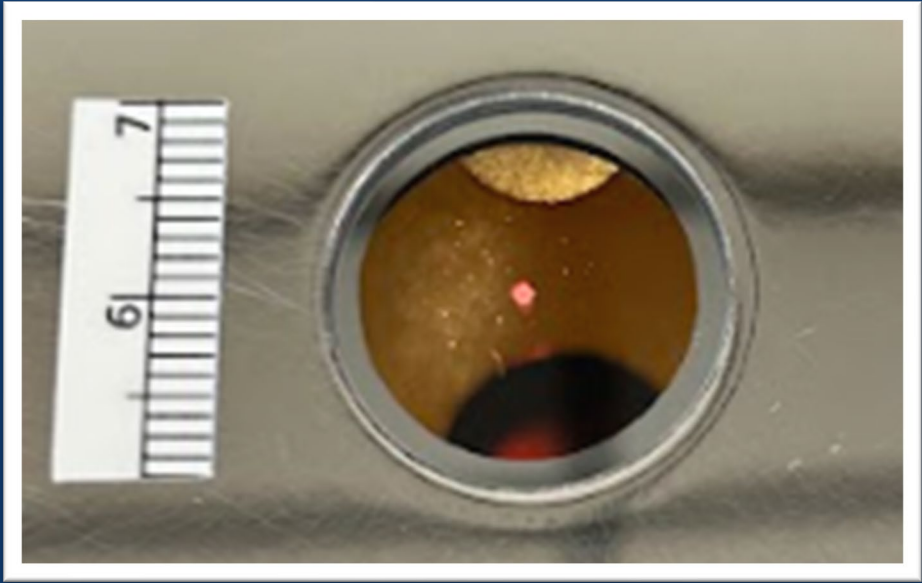
- Lifespan less than one year
- Aged via daily growth increments on statoliths; few trained experts worldwide, few sample sets aged due to expense
- Extremely variable birth, growth, and maturity rates. This makes them more vulnerable to climate-driven changes.
- Assessments infrequent, life history poorly understood and populations difficult to track due to environmental influence



Goal of this project:

Proof of concept study to evaluate feasibility of using FT-NIRS to age squid using statoliths

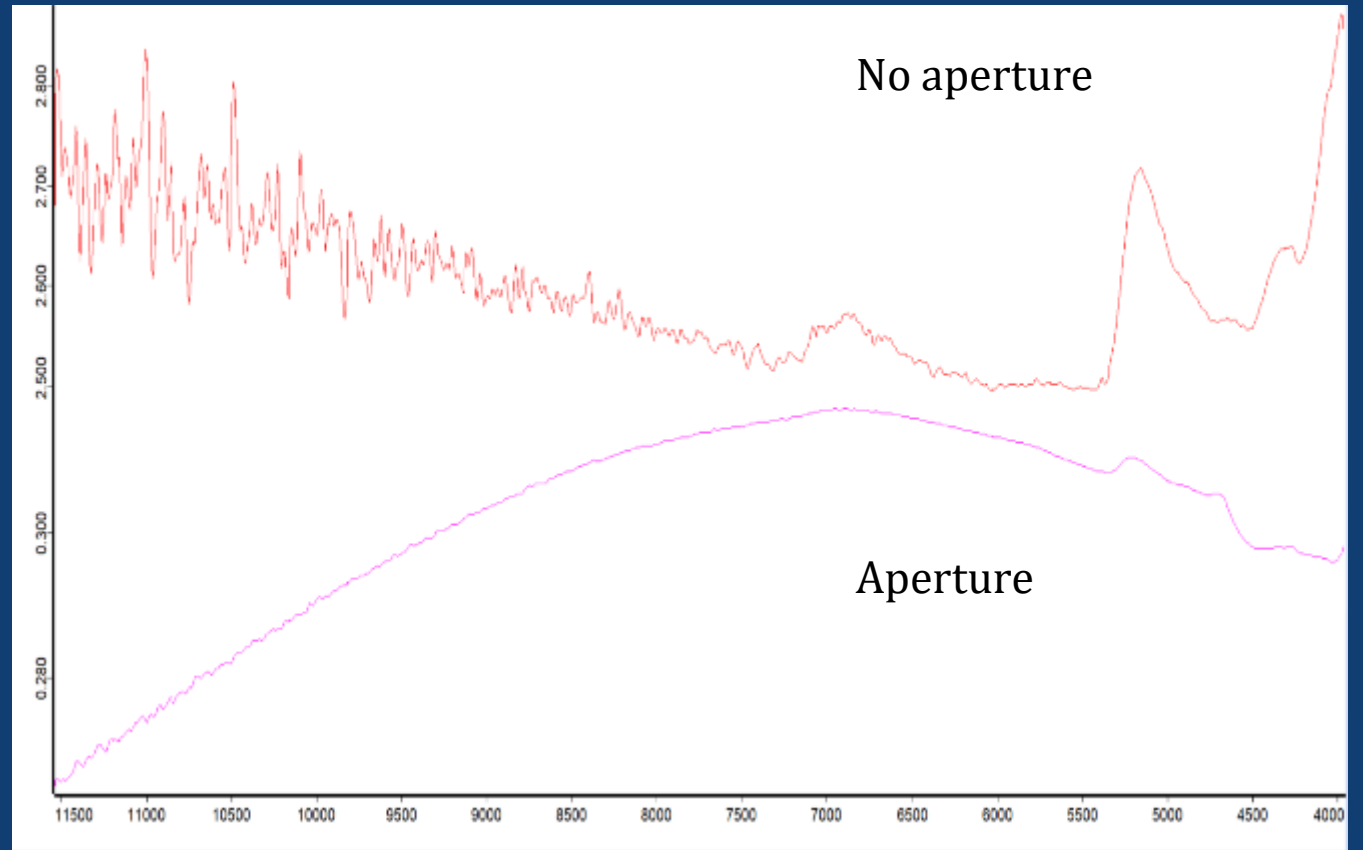
Statolith presentation



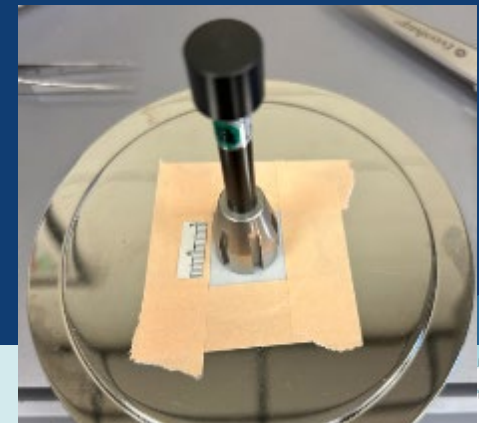
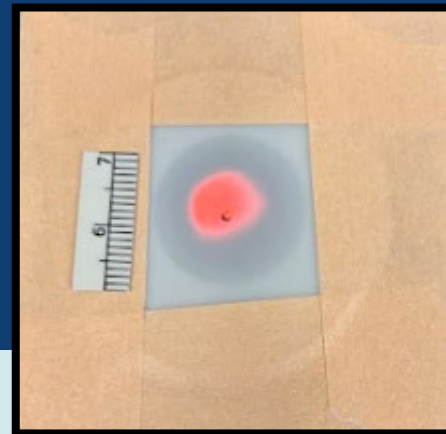
Teflon aperture to improve signal to noise



Juvenile red snapper otolith



Beer's Law becomes non-linear at amplitudes > 2



Methods



FT-NIRS model development

- Scanned at resolution of 16 cm^{-1} , 64 scans averaged to final spectrogram ($\sim 60\text{ sec}$)
- Model: Partial least squares (PLS) regression, first derivative/SNV preprocessing, wavelength selection
- Cross validated within the calibration data set using “leave-one-out” method (OPUS Software, Bruker Scientific)

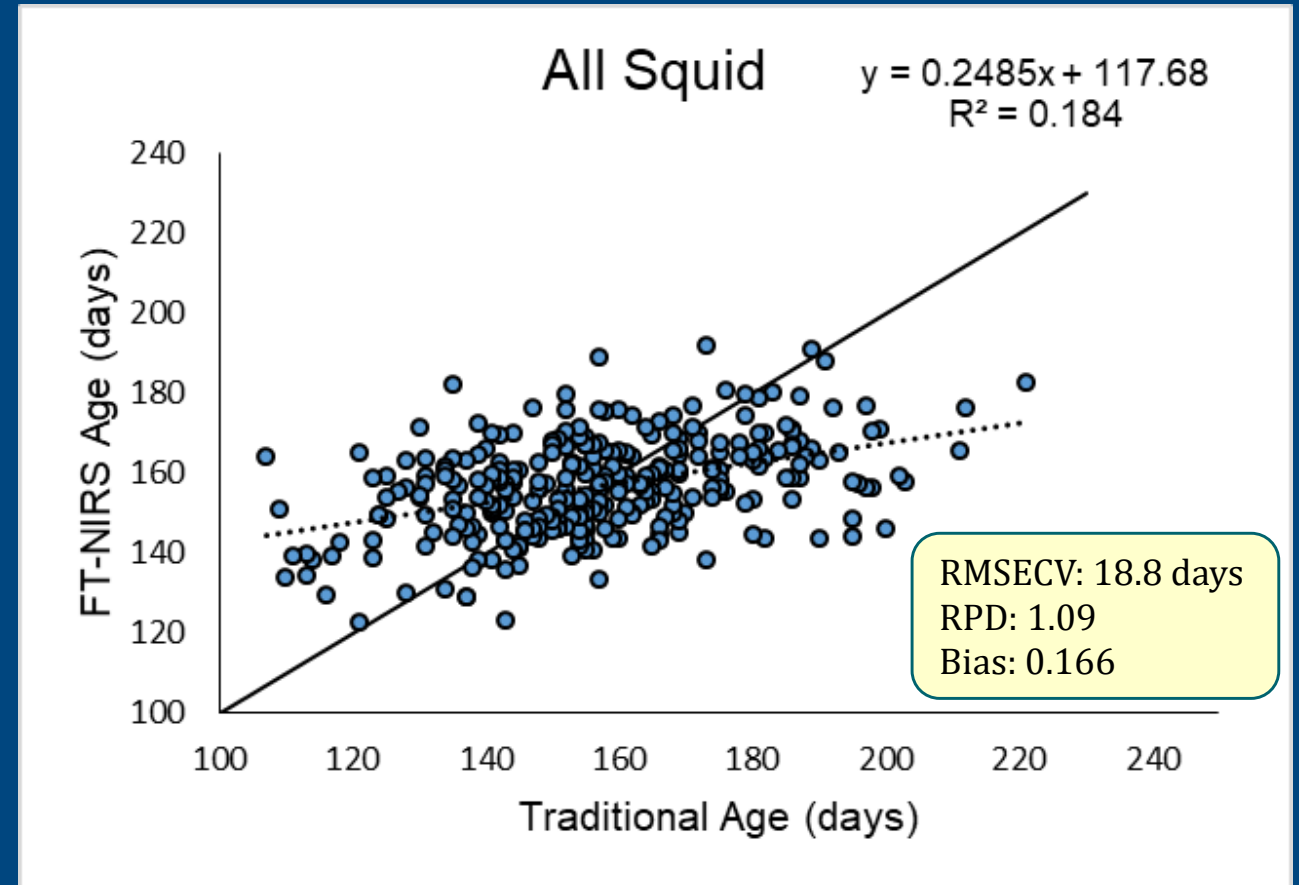
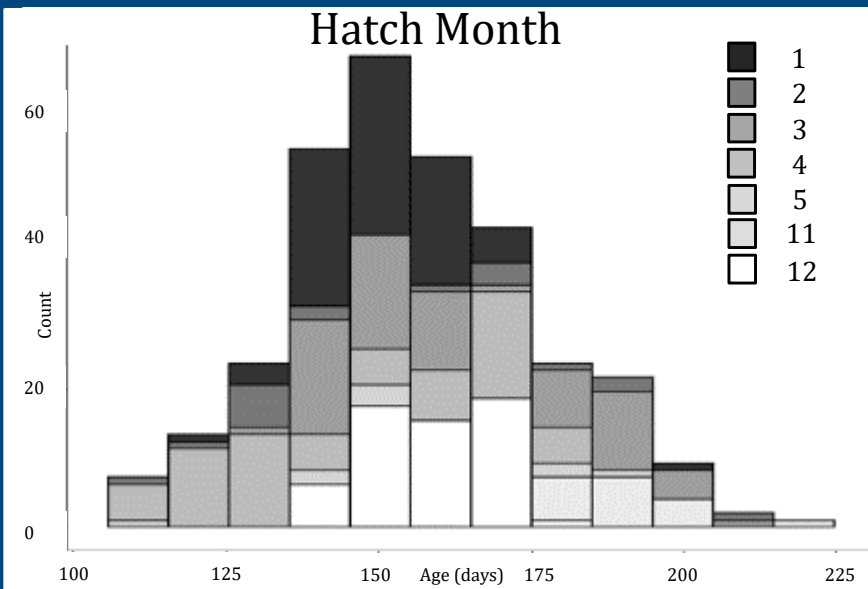
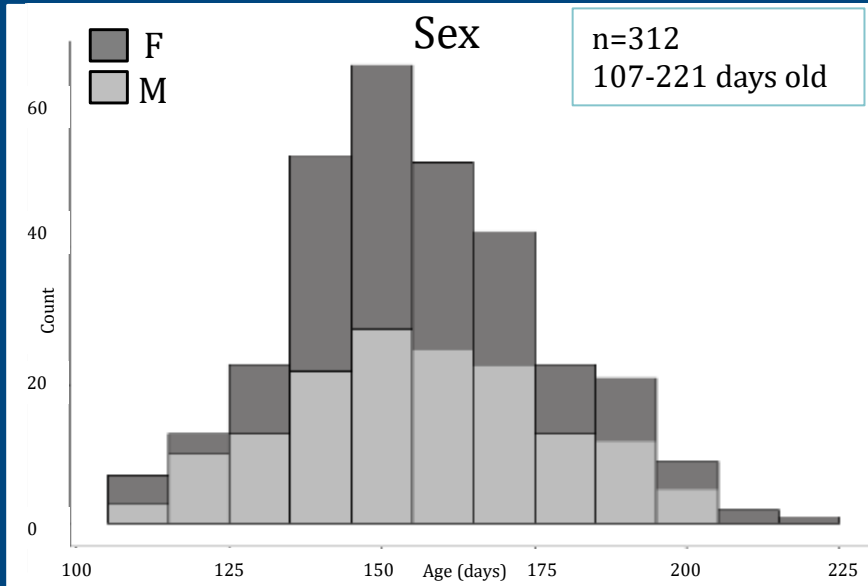
Evaluation metrics:

R^2 – coefficient of determination (maximized)

RMSECV – Root mean square error of cross-validation (minimized)

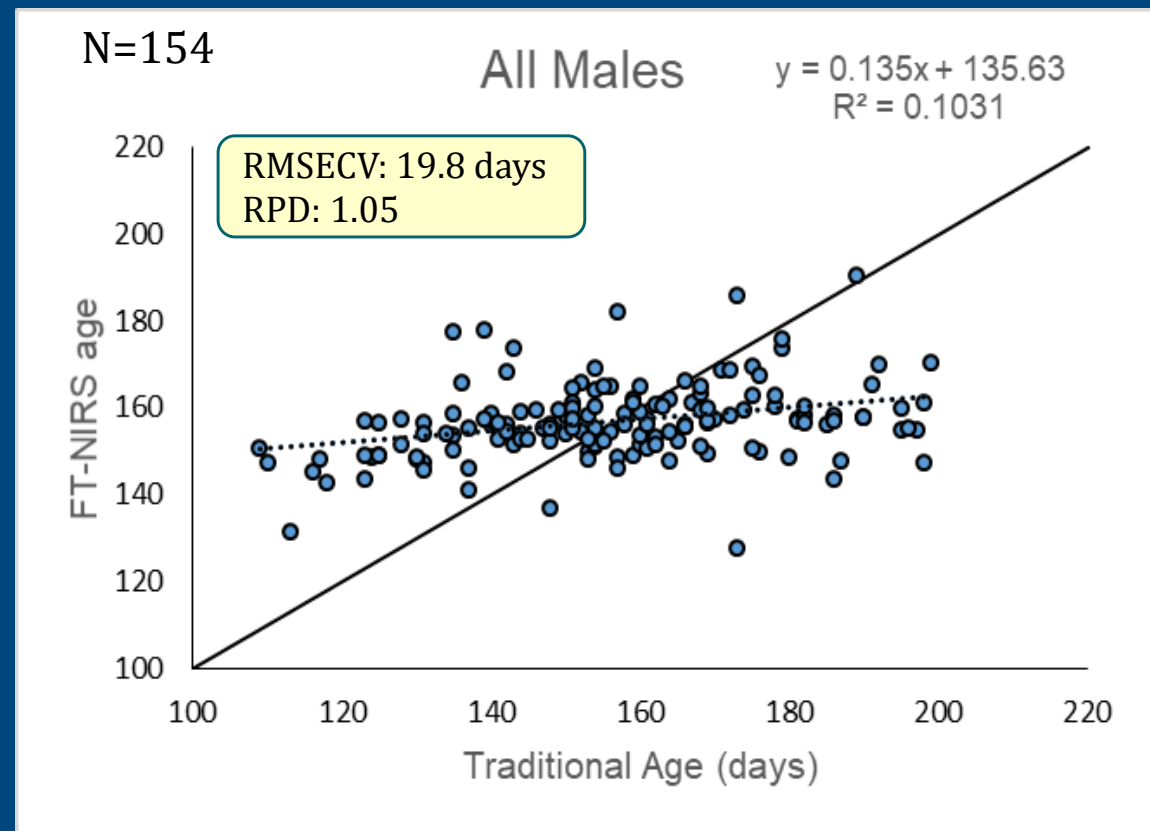
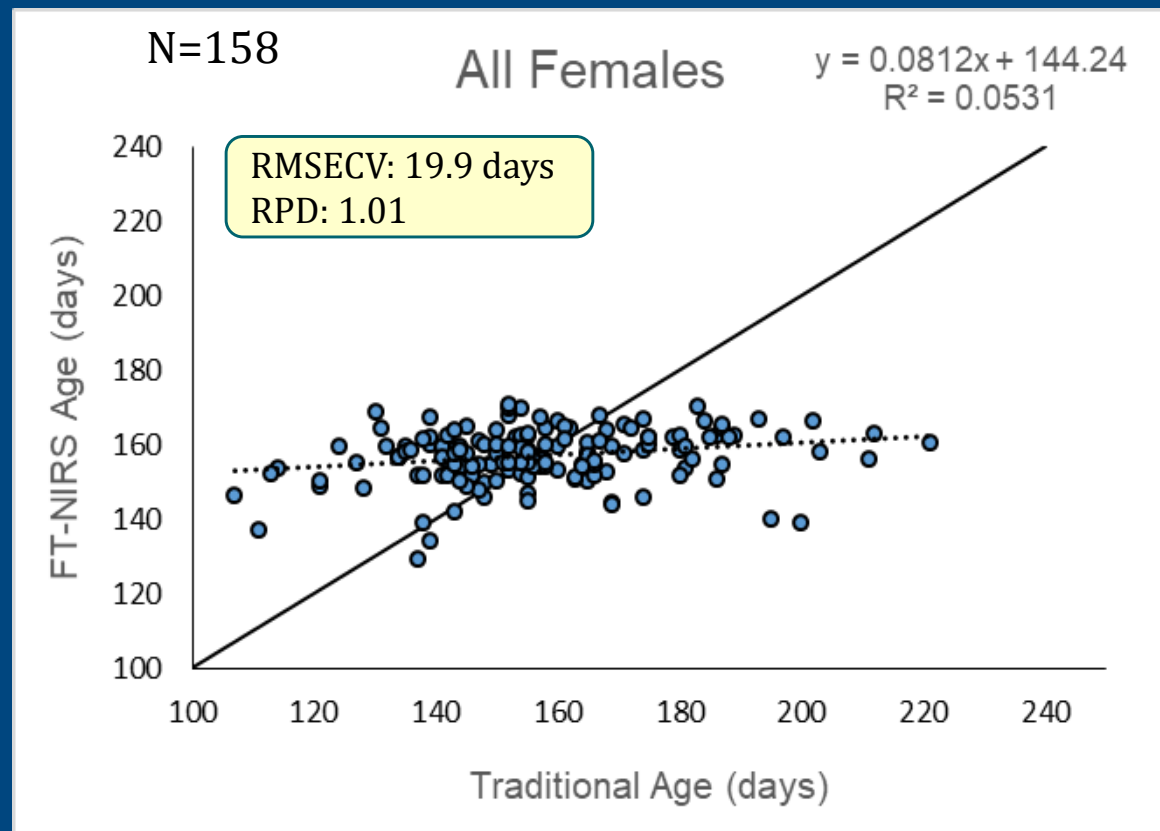
RPD – Residual prediction deviation (≥ 3 for robust models)

Results: Full cross-validation model

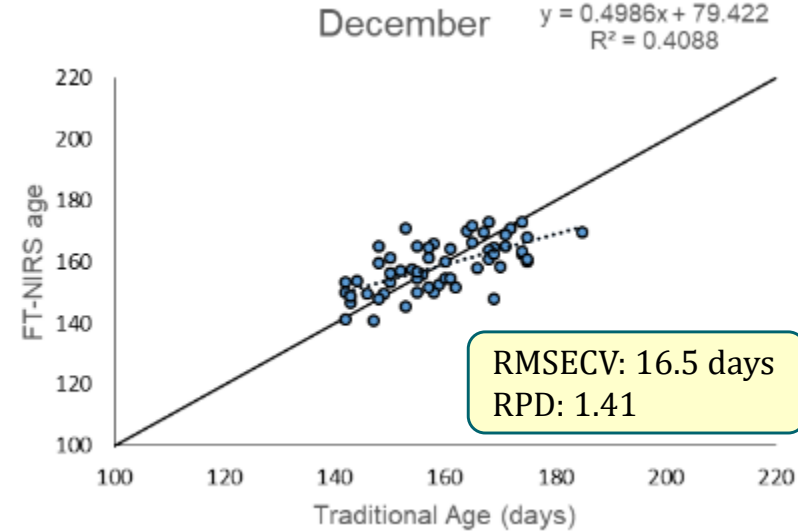
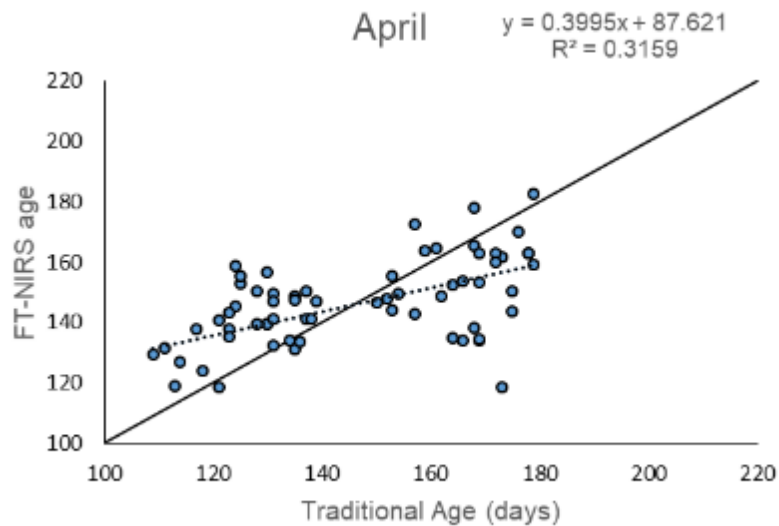
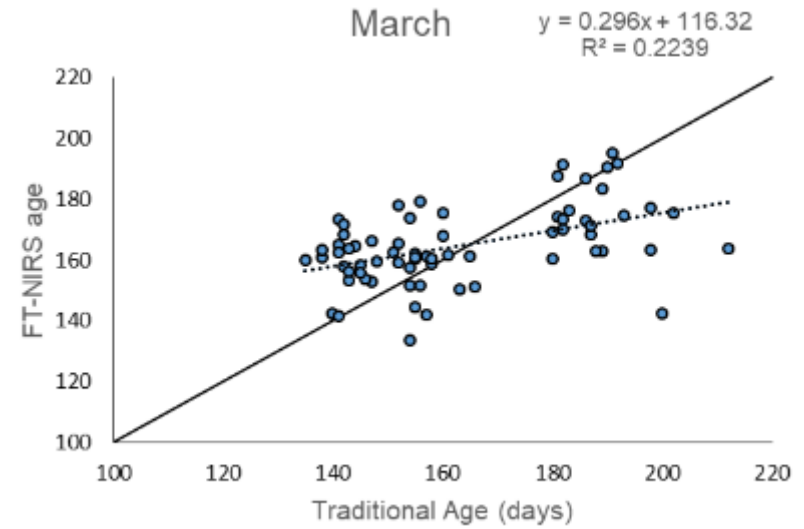
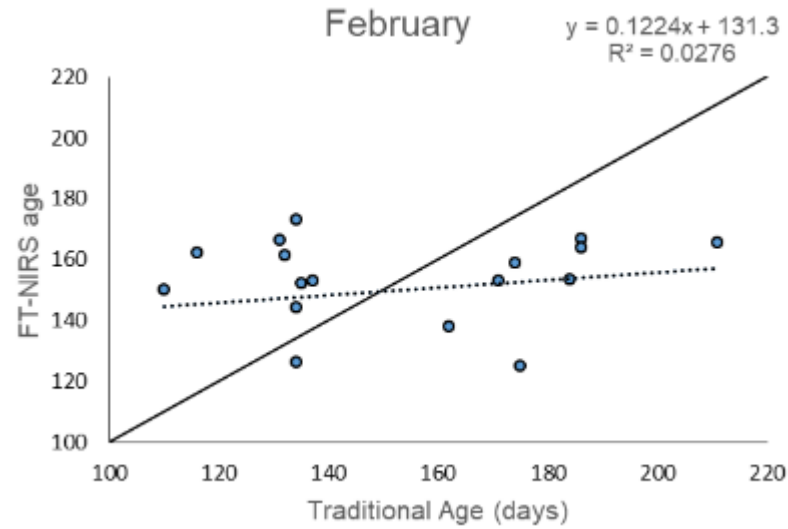
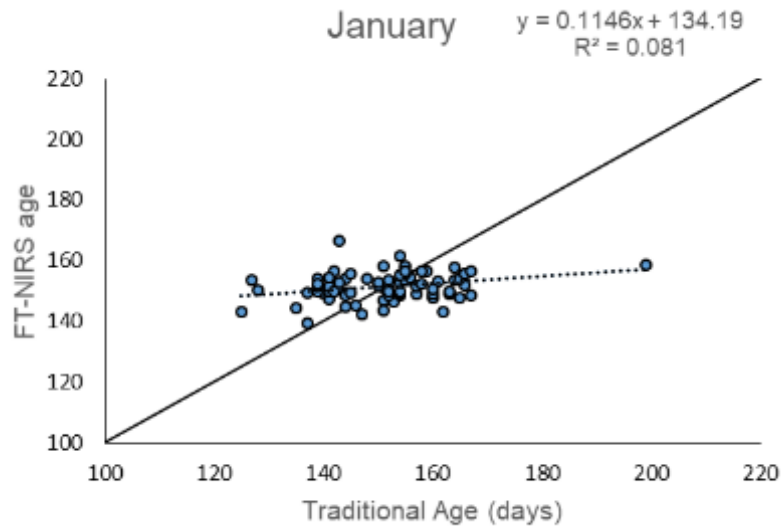


Need to consider biological factors of sex and hatch month?

Results: Sex-specific cross-validation model

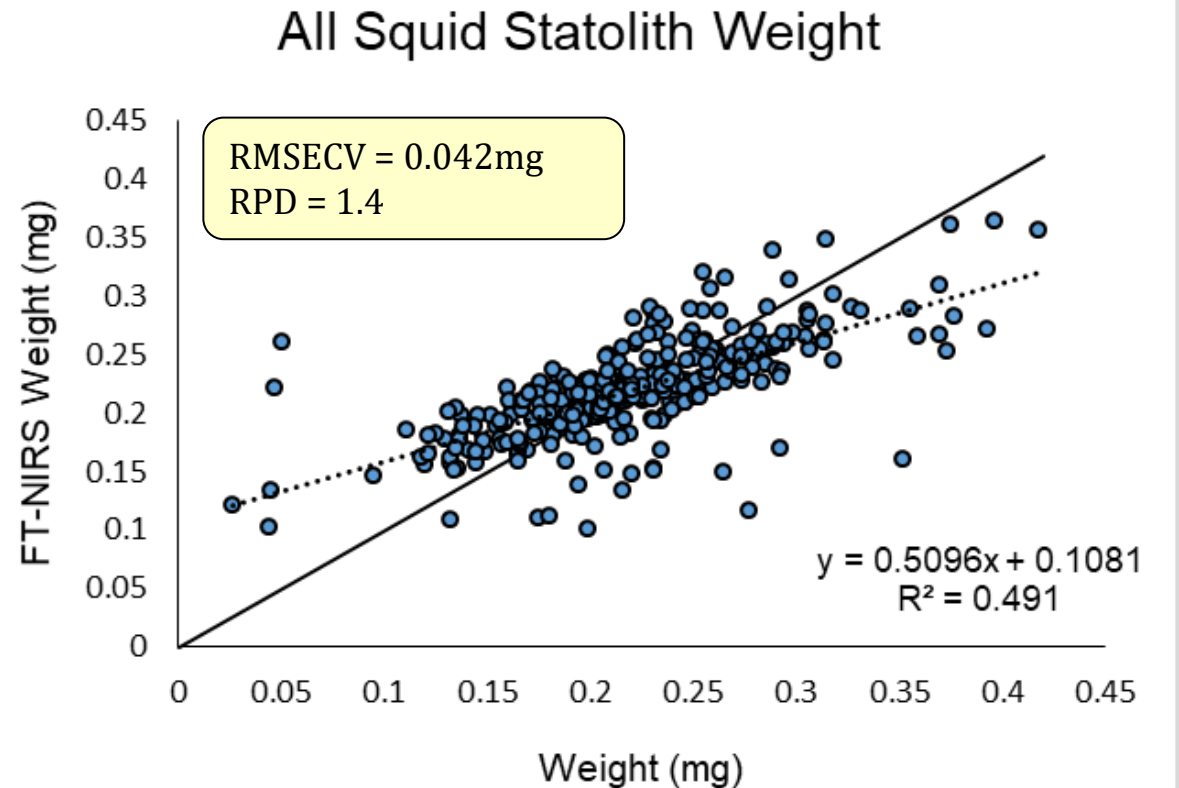
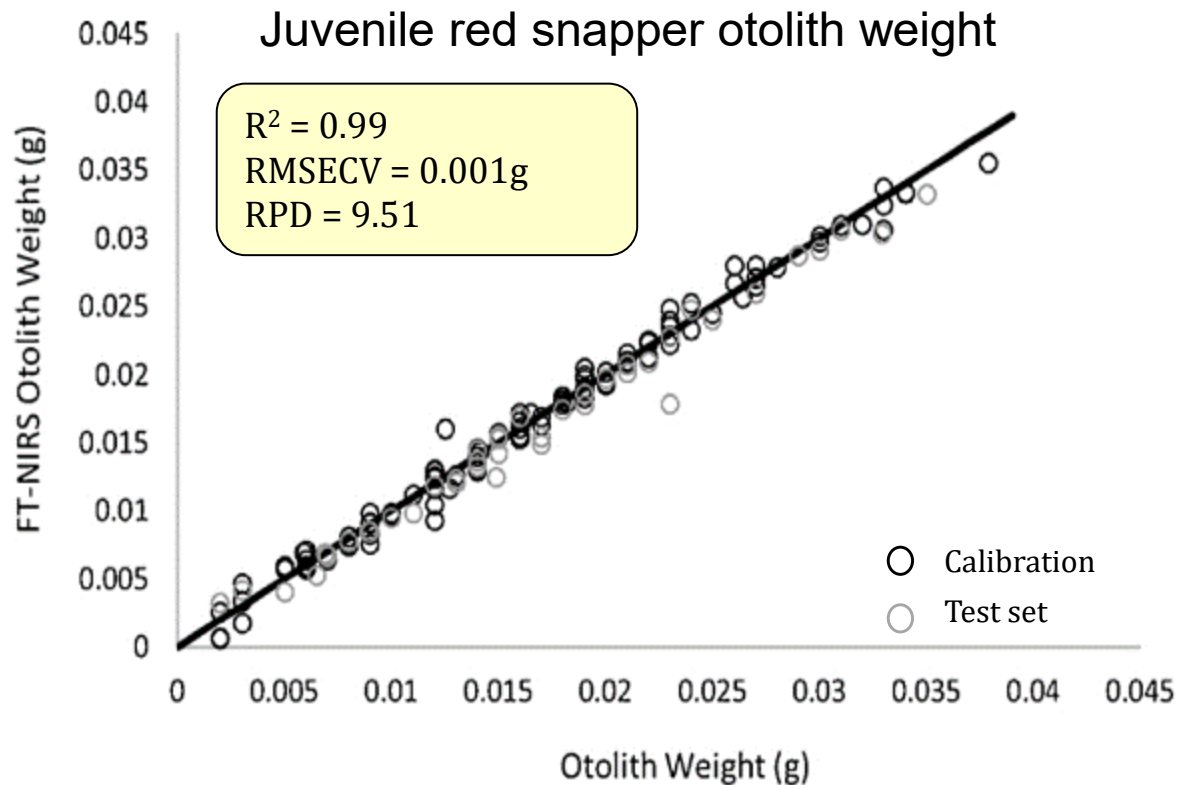


Results: Hatch month-specific cross-validation model



RMSECV: 16.5 days
RPD: 1.41

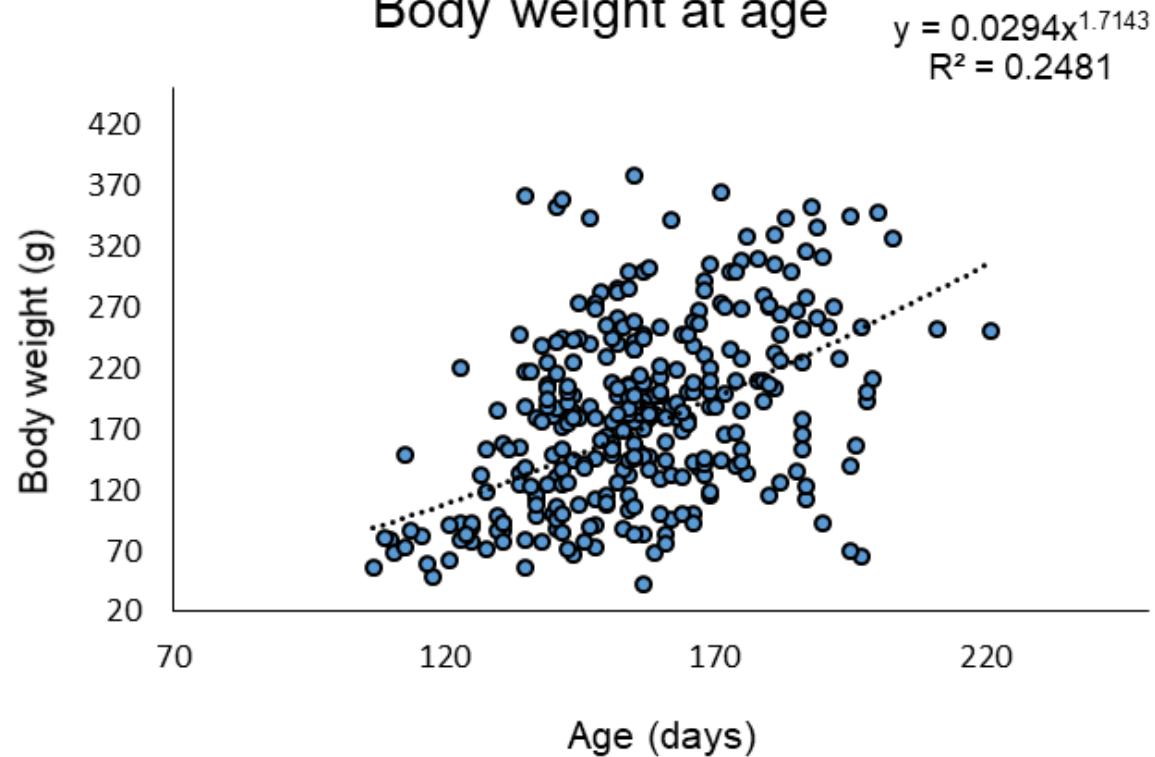
Results: Size prediction



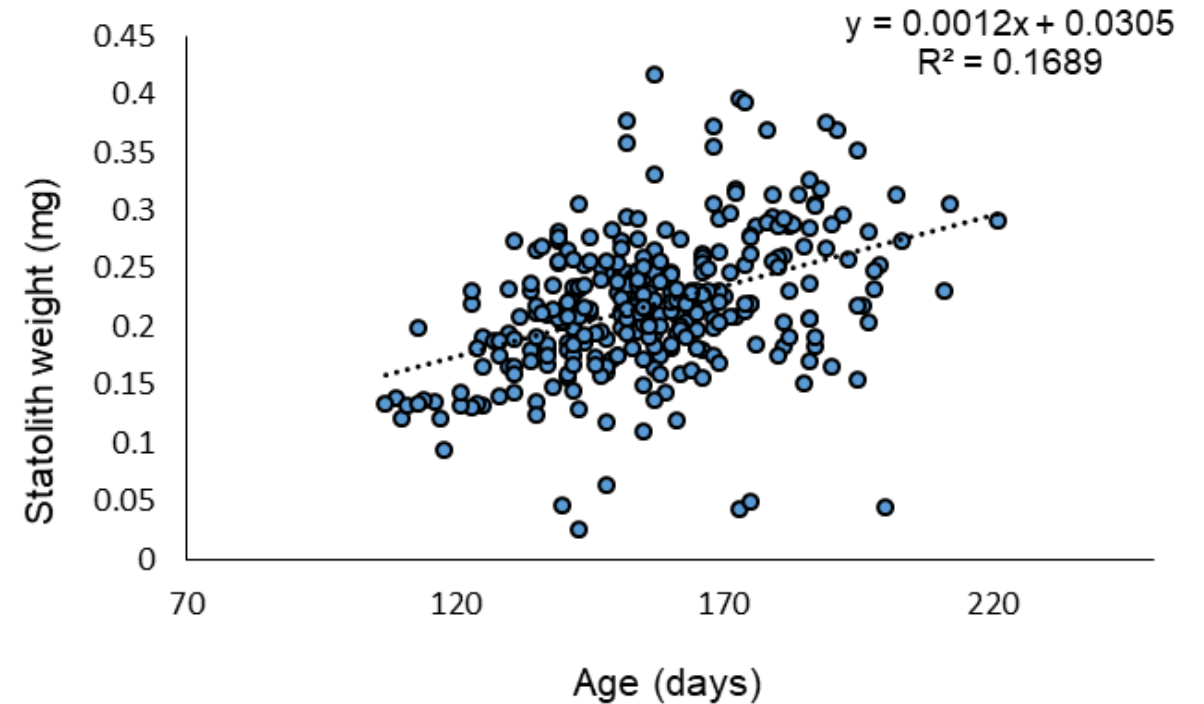
Passerotti, M. S., Jones, C. M., Swanson, C. E., & Quattro, J. M.
(2020). Fisheries Research, 223, 105439.

What's going on here?

Body weight at age



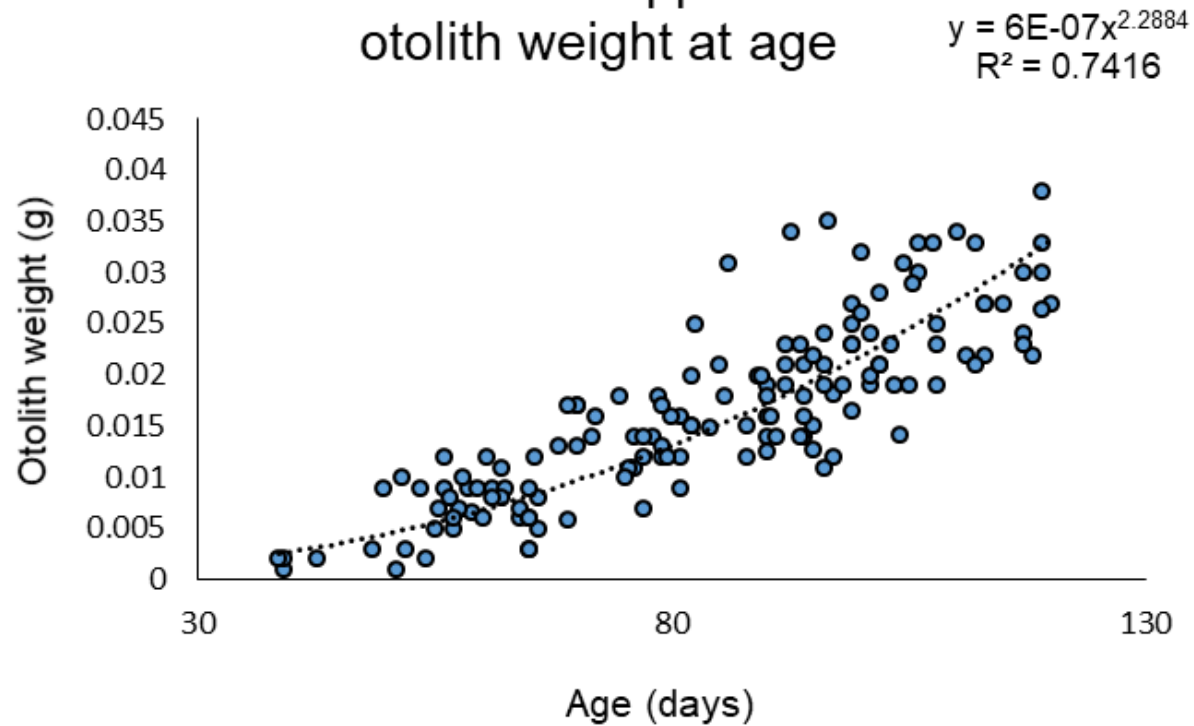
Statolith weight at age



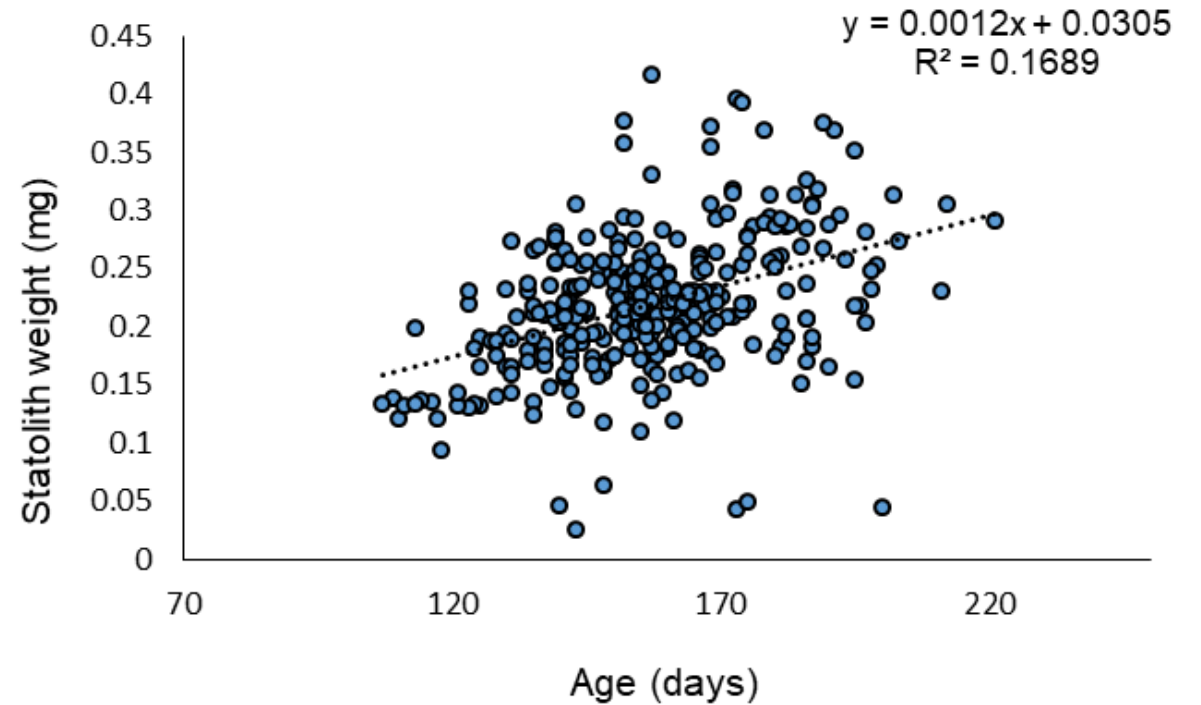
Growth plasticity creates a difficult system to model

What's going on here?

Red snapper
otolith weight at age



Statolith weight at age



Passerotti, unpublished

Conclusions: Case study #2

- No apparent capability for age prediction using FT-NIRS
 - Why?
 - Signal issue? Not if other variables can be predicted
 - Age issue? Method validated via aquarium trials
 - Growth/life history issue? Most likely
- Growth plasticity makes predicting age (and managing squid stocks) challenging
- Predictable change in the structure is necessary for FT-NIRS to be a workable method

An evaluation of the accuracy and precision of predicted ages from FT-NIR spectroscopy for two small pelagic species, Pacific sardine, *Sardinops sagax*, and Pacific mackerel, *Scomber japonicus*

Emmanis Dorval^{A,*}  and Brittany Schwartzkopf^B 

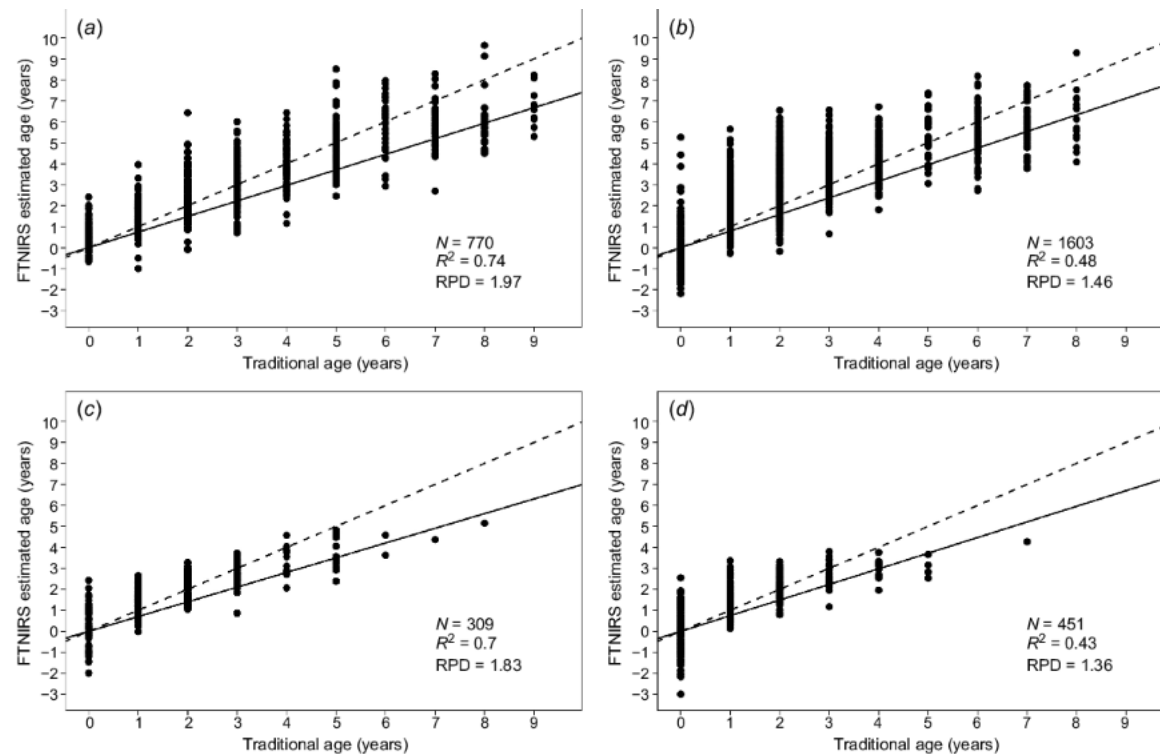


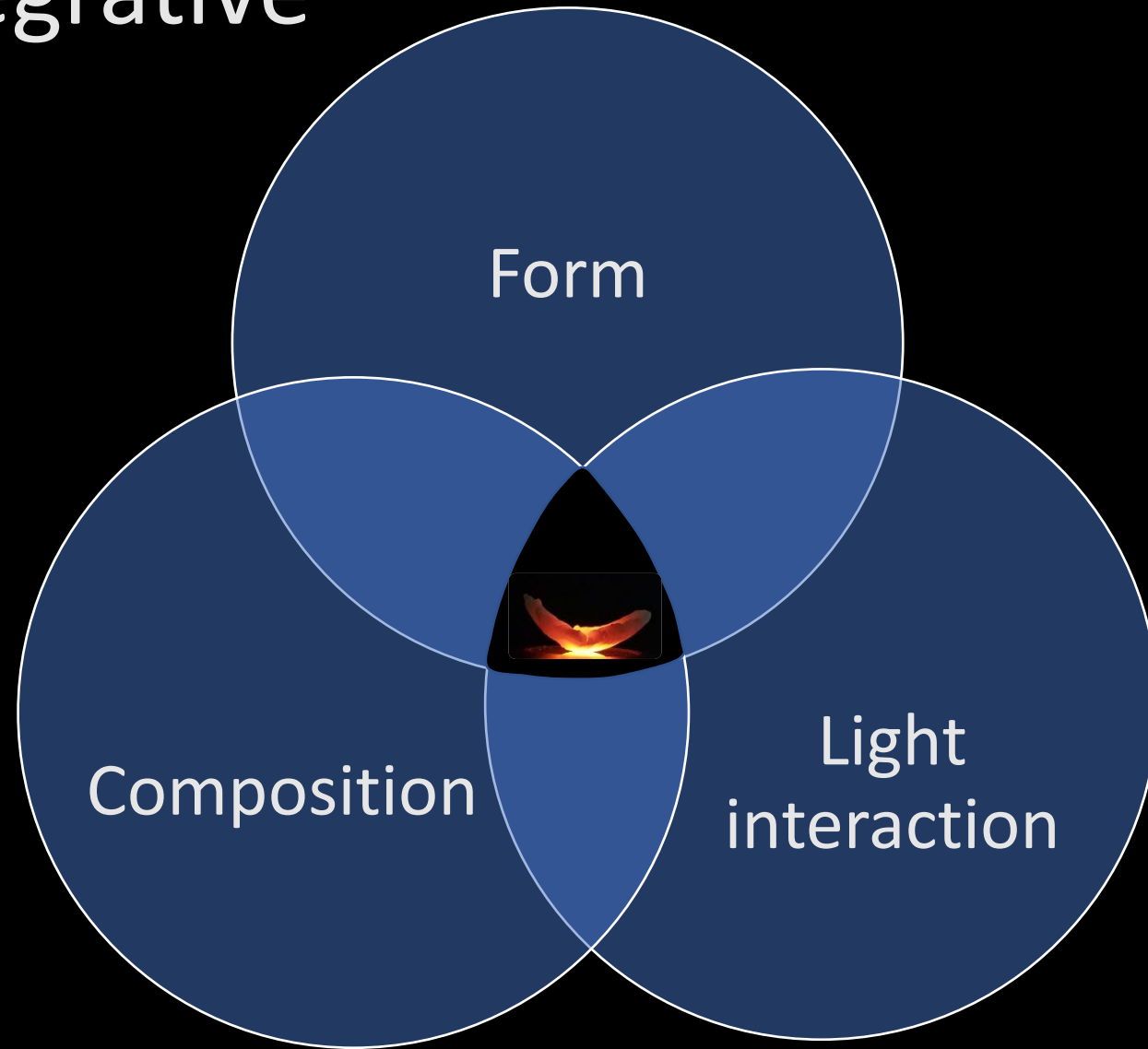
Fig. 4. Parameter estimates based on partial least squares linear regression (PLS) fitted to training and test set samples of (a, b) Pacific sardine and (c, d) Pacific mackerel. The solid line represents the slope from the PLS models predicting FT-NIRS ages, whereas the dashed line shows the 1:1 relationship between traditional ages and FT-NIRS estimated ages. RPD is the residual prediction deviation. RPD values of >3, >2, and <2 indicate good, moderate, and poor prediction respectively.

Small, short lived pelagic species also exhibit high variability in growth rates, difficulty in FT-NIRS age prediction after rapid growth tapers off

Overall Conclusions

- Common theme: sometimes FT-NIRS predictions track structural growth more closely than age
- In species with lots of variation in size at age over the lifespan, or very little change in structural size over time, this can create problems with FT-NIRS age predictions
- Indication that growth must be predictable relative to age in order to produce good age predictions

NIRS is integrative



Questions?



michelle.passerotti@noaa.gov



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